Multiple Linear Regression

A regression model that involves more than one regressor variable is called **multiple regression model**

The response y may be related to k regressor or **predictor variables**. The model

$$y = \beta_0 + \beta_1 x_1 + ... + \beta_k x_k + \epsilon$$

is called a multiple linear regression with k regressors.

The parameters $\beta_i, j = 0, 1, 2, ..., k$ are called the **regression coefficients**

The parameters β_j represents the expected change in the response y per unit change in x_j when all of the remaining regressors $x_i, (i \neq j)$ are held constant.

Estimation of Model Paramters

Least-Squares Estimation of Regression Coefficients

The method of least squares can be used to estimate the regression coefficients. Suppose that $n > k$ observations are available and let y_i denote the i^{th} observed response and x_{ij} denote the i^{th} observation or level of regressor x_j. The data will as follows:

Observation		Regressors			
i	Response, y	x_1	x_2	....	x_k
1	y_1	x_{11}	x_{12}	...	x_{1k}
2	y_2	x_{21}	x_{22}	...	x_{2k}
..	...	...	...	...	...
n	y_n	x_{n1}	x_{n2}	...	x_{nk}

We may write the sample regression model as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_k x_{ik} + \epsilon_i$$

It is more convenient to deal with multiple regression if they are expressed in matrix notation.

In matrix notation the model is

$$\mathbf{y} = \mathbf{X}\beta + \epsilon \tag{13}$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

In general, $\mathbf{y}$ is an $n \times 1$ vector of the observations, $\mathbf{X}$ is an $n \times p$ matrix of the level of regressor variables, β is a $p \times 1$ vector of the regression coefficients and ε is an $n \times 1$ vector of random errors.

We wish to find the vector of least-squares estimators $\hat{\beta}$ that minimizes:

$$S(\beta) = \sum_{i=1}^{n} \varepsilon_i^2 = \varepsilon'\varepsilon = (y - X\beta)'(y - X\beta)$$

Note that $S(\beta)$ may be expressed as

$$\begin{aligned} S(\beta) &= y'y - \beta'y - y'X\beta + \beta'X'X\beta \\ &= y'y - 2\beta'X'y + \beta'X'\beta \end{aligned} \tag{14}$$

since $\beta'X'y$ is a $1'1$ matrix or scalar and its transpose $(\beta'X'y)' = y'X\beta$ is the same scalar.

The least squares estimators must satisfy

$$\frac{\partial S}{\partial \beta} = -2X'y + 2X'X\beta = 0$$

which simplifies to:

$$X'X\hat{\beta} = X'y$$

The least squares estimator of β is

$$\hat{\beta} = (X'X)^{-1}X'y$$

provided that the inverse matrix $(X'X)^{-1}$ exists.

The fitted regression model corresponding to the levels of the regressor variables $x' = [1, x_1, x_2, ..., x_k]$ is

$$\hat{y} = x'\hat{\beta} = \hat{\beta}_0 + \sum_{i=1}^{n} \hat{\beta}_j x_j$$

The vector of fitted values $\hat{y}_i$ corresponding to the observed values y_i is

$$\hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = Hy$$

The $n \times n$ matrix $H = X(X'X)^{-1}$ is usually called the hat matrix.

The difference between the observed value y_i and the corresponding fitted value $\hat{y}_i$ is the residual $e_i = y_i - \hat{y}_i$

In terms of matrices the vector of residuals is given as **e**

$$e = y - X\hat{\beta} = y - Hy = (1 - H)y$$

Example

Fit the multiple linear regression model:

Observation	y	x_1	x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
4	14.88	4	80
5	13.75	6	150

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

to the above data.

Solution

To fit the multiple regression we first form the model X and y vector

$$X = \begin{bmatrix} 1 & 7 & 560 \\ 1 & 3 & 220 \\ 1 & 3 & 340 \\ 1 & 4 & 80 \\ 1 & 6 & 150 \end{bmatrix}, y = \begin{bmatrix} 16.68 \\ 11.50 \\ 12.03 \\ 14.88 \\ 13.75 \end{bmatrix}$$

The $X'X$ matrix is given by

$$X'X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 7 & 3 & 3 & 4 & 6 \\ 560 & 220 & 340 & 80 & 150 \end{bmatrix} \begin{bmatrix} 1 & 7 & 560 \\ 1 & 3 & 220 \\ 1 & 3 & 340 \\ 1 & 4 & 80 \\ 1 & 6 & 150 \end{bmatrix} = \begin{bmatrix} 5 & 23 & 1350 \\ 23 & 119 & 6820 \\ 1350 & 6820 & 506500 \end{bmatrix}$$

and the $X'y$ vector is

$$X'y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 7 & 3 & 3 & 4 & 6 \\ 560 & 220 & 340 & 80 & 150 \end{bmatrix} \begin{bmatrix} 16.68 \\ 11.50 \\ 12.03 \\ 14.88 \\ 13.75 \end{bmatrix} = \begin{bmatrix} 68.84 \\ 329.37 \\ 19213.9 \end{bmatrix}$$

The least squares estimator of β is

$$\hat{\beta} = (X'X)^{-1} X^{-1} Y$$

or

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 5 & 23 & 1350 \\ 23 & 119 & 6820 \\ 1350 & 6820 & 506500 \end{bmatrix}^{-1} \begin{bmatrix} 68.84 \\ 329.37 \\ 19213.9 \end{bmatrix} = \begin{bmatrix} 1.832 & -0.325 & -0.000504 \\ -0.325 & 0.0945 & -0.000406 \\ -0.0005 & -0.0004 & 0.00000878 \end{bmatrix} \begin{bmatrix} 68.84 \\ 329.37 \\ 19213.9 \end{bmatrix} = \begin{bmatrix} 9.32 \\ 0.946 \\ 0.00035 \end{bmatrix}$$

The model is given by:

$$Y = 9.32 + 0.946x_1 + 0.00035x_2$$

The above example can be done in R using:

```r
y <- c(16.68,11.50,12.03,14.88,13.75)
x1 <- c(7,3,3,4,6)
x2 <- c(560,220,340,80,150)
model <- lm(y ~ x1+ x2)
knitr::kable(broom::tidy(model))
```

term	estimate	std.error	statistic	p.value
(Intercept)	9.3200051	2.2722506	4.1016624	0.0546162
x1	0.9463629	0.5161297	1.8335758	0.2081636
x2	0.0003508	0.0049762	0.0705021	0.9502094

Properties of the Least-Squares Estimators

The least squares estimator $\hat{\beta}$ is unbiased:

$$E(\hat{\beta}) = E\Big[(X'X)^{-1}X'Y\Big] = E\Big[(X'X)^{-1}X'X'(X\beta+\varepsilon)\Big] = E\Big[(X'X)^{-1}X'X\beta+(X'X)^{-1}X'\varepsilon\Big] = \beta$$

since $E(\varepsilon) = 0$ and $(X'X)^{-1}X'X = I$. Thus $\hat{\beta}$ is unbiased estimator of β if the model is correct.

The variance of $\hat{\beta}$ is given by:

$$Var(\hat{\beta}) = Var\Big[(X'X)^{-1}X'y\Big] = (X'X)^{-1}X'Var(y)\Big[(X'X)^{-1}X'\Big] = \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X$$

Estimation of σ^2

We devleop the estimator of σ^2 from the residual sum of squares

$$SS_{Res} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} e_i^2 = e'e$$

Replacing

$$e = y - X\hat{\beta}$$

we have

$$\begin{aligned} SS_{Res} &= (y - X\hat{\beta})'(Y - X\hat{\beta}) \\ &= y'y - \hat{\beta}X'y - y'X\hat{\beta} + \hat{\beta}X'X\hat{\beta} \\ &= y'y - 2\hat{\beta}'X'y + \hat{\beta}'X'X\hat{\beta} \end{aligned} \tag{15}$$

Since $X'X\hat{\beta} = X'y$$ this last equation becomes

$$SS_{Res} = y'y - \hat{\beta}'X'Y$$

The residual sum of squares has $n-p$ degrees of freedom associated with it since p parameters are estimated in the regression model.

The **residual mean square** is

$$MS_{Res} = \frac{SS_{Res}}{n-p}$$

Maximum Likelihood Estimation

The model is given by:

$$Y = X\beta + \varepsilon$$

and the errors are normally and independently distributed with constant variance σ^2 or ε is distributed as $N(0, \sigma^2 I)$.

The normal denisity function for the error is

$$f(\varepsilon_i) = \frac{1}{\sigma\sqrt{2\pi}} \text{exp}\left(-\frac{1}{2\sigma^2}\varepsilon_i^2\right)$$

The likelihood function is the joint density of $\varepsilon_1, \varepsilon_2, ..., \varepsilon_n$ or $\prod_{i=1}^{n} f(\varepsilon_i)$. Therefore the likelihood function is:

$$L(\varepsilon, \beta, \sigma^2) = \prod_{i=1}^{n} f(\varepsilon_i) = \frac{1}{(2\pi)^{n/2}\sigma^n} exp\Big(-\frac{1}{2\sigma^2}\varepsilon'\varepsilon\Big)$$

Now since we can write $\varepsilon = y - X\beta$ the likelihood function becomes

$$L(y, X, \beta, \sigma^2) = \frac{1}{(2\pi)^{n/2}\sigma^n} exp\Big(-\frac{1}{2\sigma^2}(y - x\beta)'(y - x\beta)\Big)$$

The log-likelihood function becomes

$$\ln L(y, X, \beta, \sigma^2) = -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)$$

The maximum likelihood estimator of β under normal errors is equivalent to the least squares estimator $\hat{\beta} = (X'X)^{-1}X'y$

The maximum likelihood estimator of σ^2 is

$$\hat{\sigma^2} = \frac{(Y - x\hat{\beta})'(y - X\hat{\beta})}{n}$$